AD 502 – Machine Learning

Pre-Reading Material – Lecture 14

Decision Tree

Estimated Reading Time: 10–15 minutes

Why This Topic Matters

A **Decision Tree** is a supervised machine-learning algorithm that makes predictions through a sequence of simple **if–then decisions**. It is widely used because its decision-making process is easy to visualize and interpret.

For example:

Study Hours > 4? → Yes → Attendance > 75%? → Yes → Pass

Decision Trees can be used for both **classification and regression**.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the basic structure of a Decision Tree.
- Identify root, internal, branch, and leaf nodes.
- Understand **entropy, information gain, and Gini index**.
- Explain how the best split is selected.
- Understand overfitting and pruning.

Core Concepts

1. Decision Tree Structure

A Decision Tree consists of:

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- **Root Node:** First decision/split.
- **Internal Node:** Intermediate decision.
- **Branch:** Outcome of a decision.
- **Leaf Node:** Final prediction.

2. How Does It Choose a Split?

The tree searches for the feature that produces the **best separation of classes**.

Two common measures are:

Entropy:

$H(S)=-\sum p_i \log_2(p_i)$

Entropy measures the **impurity/uncertainty** of a node.

- Entropy = 0 → Completely pure
- Higher entropy → More mixed classes

Information Gain:

$IG=H(\text{parent})-\text{Weighted } H(\text{children})$

A split with **higher information gain** generally provides a better separation.

Gini Index:

$Gini=1-\sum p_i^2$

Lower Gini → **Purer node**.

Example

Suppose we want to predict whether a student will **Pass or Fail**.

Study Hours Attendance Result

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Study Hours	Attendance	Result
6	90%	Pass
5	85%	Pass
2	60%	Fail
3	65%	Fail

The Decision Tree may learn:

Study Hours > 4?

→ Yes → Pass

→ No → Fail

The algorithm mathematically evaluates possible splits and selects the one that reduces impurity most effectively.

Overfitting and Pruning

If a Decision Tree grows too deeply, it may **memorize the training data** rather than learn general patterns.

This is called **overfitting**.

To control overfitting, we use **pruning**—removing unnecessary branches.

- **Pre-pruning:** Stop the tree from growing beyond certain limits.
- **Post-pruning:** Grow the tree and then remove unnecessary branches.

Think Before the Lecture

1. Why should a Decision Tree not simply keep growing until every training example is classified correctly?

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2. If two features can split the data, how can we mathematically decide which one is better?
3. Why does a pure node have zero entropy and zero Gini impurity?
4. What is the difference between **classification** and **regression** using Decision Trees?

Key Takeaway

A Decision Tree learns a sequence of decision rules by selecting splits that make the resulting groups as pure as possible.

The key concepts for Lecture 4 are:

Decision Tree → Split → Entropy/Gini → Information Gain → Leaf → Prediction → Overfitting → Pruning

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